Task F1 — Three Model Variants

FP32 baseline, PTQ INT8, and structured pruning + PTQ INT8 | LogiEdge | FreightBridge Logistics Pvt. Ltd. | Component F, Module 6

At a glance

Item	Result
Variants	M1 FP32 baseline 3,212 B of numbers; M2 PTQ INT8 1,568 B; M3 35% pruned + PTQ INT8, network 6→21→10→3, 400 parameters (−50%)
Calibration	234 windows for the INT8 ranges (at least 200 required)
Status	**The TensorFlow / tfmot build has not been run** (neither library was available to me). All three variants were built and measured in numpy with TFLite's full-integer arithmetic; `build_variants_tf.py` makes the real .tflite files.
Baseline	M1 is the Task D1 recommended model (trained with mid-run-onset runs). The required D1 model reaches only 94.1% Critical recall on the 57-window split, one window short of the 95% target in Task F3.

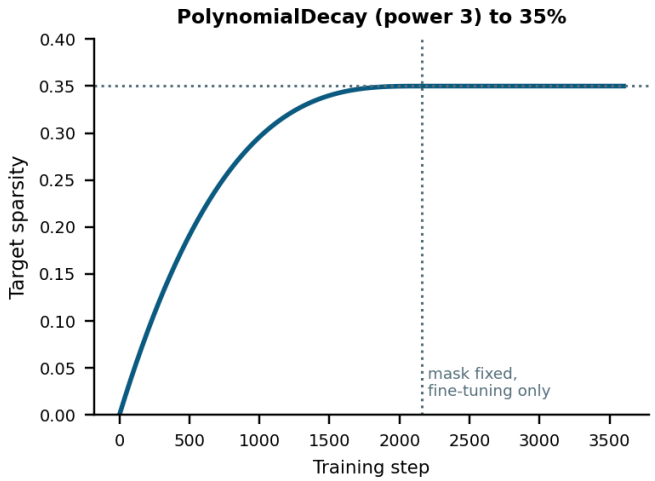
1 The three variants

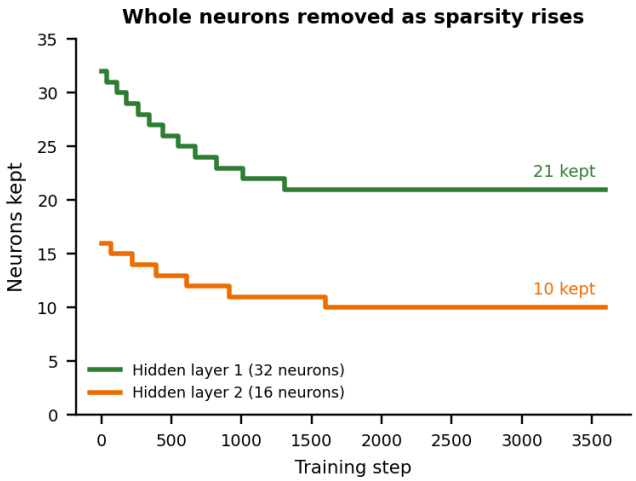
Variant	Method	Network	Parameters	Stored numbers	Tools
M1	Standard training (Task D1), FP32	6 → 32 → 16 → 3	803	3,212 B (FP32)	tf.keras (numpy here)
M2	Post-training quantisation, full INT8, calibrated on 234 windows	6 → 32 → 16 → 3	803	1,568 B (INT8 weights, INT32 biases, scales)	TFLiteConverter, DEFAULT + representative dataset
M3	35% structured pruning (PolynomialDecay), fine-tune, remove neurons, then full INT8 PTQ	6 → 21 → 10 → 3	400	910 B	tensorflow_model_optimization + TFLiteConverter

2 M2: post-training INT8 quantisation

- **Scheme (as TFLite's full-integer mode):** int8 activations with a scale and zero point from the min and max seen on the calibration windows; symmetric int8 weights per output channel; int32 biases; int32 accumulation; requantisation to int8 after every layer.
- **Calibration:** the 234 training windows (the code refuses fewer than 200).
- **One thing to know:** the normalised inputs run from about −3 (Normal) to about 1,300 (a 70 °C window), so the input scale is 4.99, about five Normal-class σ per step. The whole Normal range lands on one or two integer levels. Accuracy still holds on clean data (Task F2), but it makes INT8 less tolerant of noise.
- **Per channel or per tensor:** both gave identical predictions here. Newer TensorFlow versions quantise Dense layers per channel by default; whichever your version uses, the results should match.

3 M3: structured pruning with a polynomial-decay schedule





Step	What happens
Granularity	Whole hidden neurons (the “filters” of a dense layer). In tfmot this is `block_size=(inputs, 1)`: one whole kernel column is pruned at a time. Unstructured pruning would zero single weights and leave the file the same size.
Importance	Mean absolute value of a neuron's incoming weights (tfmot's AVG block pooling)
Schedule	PolynomialDecay from 0% to 35%, power 3, reached at 60% of training; mask updated every 10 steps; output layer not pruned
Fine-tuning	200 epochs of Adam on the training windows plus the mid-run-onset runs; pruned weights stay zero
Removal	The zeroed neurons are deleted with their matching rows in the next layer: 32 → 21 and 16 → 10 neurons. A neuron's constant output (relu of its bias) is folded into the next bias, so the smaller network computes the same function (tested).
Result	803 → 400 parameters (−50%); 752 → 366 multiply-accumulates per inference (−51%)

4 The TensorFlow build (for you to run)

```
python3.11 -m venv .venv && . .venv/bin/activate
pip install tensorflow==2.14.1 tensorflow-model-optimization==0.8.0 numpy==1.26.4 psutil
python3 build_variants_tf.py          # M1 SavedModel, m1_fp32.tflite, m2_ptq_int8.tflite, m3_pruned_int8.tflite
```

Specified setting	Where it is in build_variants_tf.py
tf.lite.Optimize.DEFAULT	`converter.optimizations = [tf.lite.Optimize.DEFAULT]`
Representative dataset, 200 or more samples	234 training windows; raises an error below 200
Full INT8	`TFLITE_BUILTINS_INT8`, int8 input and output
35% structured pruning, PolynomialDecay	`PolynomialDecay(final_sparsity=0.35)` with `prune_low_magnitude(block_size=(inputs, 1))`
Checks built in	Keras M1 matches the trained weights; the physically smaller M3 matches the stripped model; each .tflite is scored on the validation set; exits with an error if Critical recall is 95% or below

5 Verification and items to confirm

Check	How	Result
INT8 helpers, ranges, calibration limit, per-channel scales, agreement with FP32	28 tests	Pass
Pruning schedule values, 21 and 10 neurons kept, neuron removal keeps the function	Tests	Pass
Stored-number counts, MACs, activation memory	Tests	Pass
TensorFlow script has the specified settings and parses	Static check	Pass
Running build_variants_tf.py, real .tflite files	Needs TensorFlow and tfmot	Not run

- **Baseline.** I used the D1 onset-trained model as M1. Tell me if you want the required D1 model instead; its Critical recall on the 57-window split is 94.1% (16 of 17), just under the Task F3 target.
- **“Filter”.** I took it to mean a hidden neuron, since the network is dense. 35% is applied to each hidden layer; the output layer keeps its 3 neurons.
- **Versions.** TensorFlow 2.14.1 with tfmot 0.8.0 (tested together) and tflite-runtime 2.14.0 match the earlier tasks. The number of neurons tfmot removes may differ from mine by one.